Design and Development of Jet Engine Model

Submitted In Partial Fulfillment of Requirements
For the Degree Of

Bachelor of Technology
(Mechanical Engineering)

By

Shivam Takle
Roll No: 16010520073

Sourish Shetty
Roll No: 16010520078

Kushal Karelia
Roll No: 16010520080

Guide

Prof. Sandeep Chopade

Somaiya Vidyavihar University
Vidyavihar, Mumbai - 400 077

2020-24

Somaiya Vidyavihar University

K. J. Somaiya College of Engineering

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This is to certify that the dissertation report entitled **Design and Development of Jet Engine Model** submitted by Shivam Takle, Sourish Shetty and Kushal Karelia at the end of semester VIII of LY B. Tech is a bona fide record for partial fulfillment of requirements for the degree Bachelor of Technology (Mechanical Engineering) of Somaiya Vidyavihar University

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_____

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Abstract

Jet engines see wide use in many applications, aviation and energy production among many others. The design and construction of a jet engine requires a great deal of knowledge from many different fields. From thermodynamics and fluid mechanics to mechanical engineering.

In order to build a modern jet engine, you need a lot of experienced people from all these certain fields. A modern jet engine is truly a miracle of engineering. With everything from fine tolerance in space to resilience to high temperatures and stress, the jet engine has gone through a revolution over the years, with great improvements in performance, efficiency and reliability. The most commonly known jet engines are the turbojet engine, the turboprop engine, the turbofan engine, the turboshaft and the ramjet engine.

The major principles in all these engines are the same. And they work according to similar concepts as the internal combustion engine: suck, squeeze, bang and blow. The first part is focused on the inlet, the air is sucked in: suck. The second part is focused on the compression of the air, where the inlet air is compressed to a higher pressure: squeeze. The third part is focused on the combustion chamber, where the compressed air is mixed with fuel and then ignited: bang. The fourth part is focused on the outlet of the engine, where the ignited air and fuel-mixture exits at a high velocity: blow. These concepts can be applied to all of the jet engines, with minor adjustments. Starting out with a compressor, a combustion chamber and a turbine, we can tell the difference from the turbojet, turboprop, turboshaft and turbofan on where the power is transferred from the turbine. In a turbojet, the turbine is connected to the compressor.

In summary, jet engines are marvels of engineering that have transformed the way we travel and explore the skies, making the world a more interconnected place. Their continuous development and innovation drive the progress of aviation and aerospace technologies.

***Key words*:** Jet Engine, Impeller, Stator, Motor, Combustion Chamber, Thrust, Flow Simulation, Fuel.

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Chapter 1: Introduction

The chapter outlines the scope, objective and purpose of the project. The groundwork laid in this introduction paves the way for a detailed exploration of the system's design methodology and its potential future enhancements.

1.1 Objective

Objective: Ensure the jet engine operates safely and reliably under various conditions.

Key Activities: Implement safety features, conduct stress tests to identify potential weaknesses, and refine the design to enhance reliability and durability.

This project aims to achieve the following objectives:

- Design and Analysis:

  Objective: Develop a detailed design for a jet engine, considering factors such as efficiency, thrust-to-weight ratio, and safety.

  Key Activities: Conduct aerodynamic and thermodynamic analysis, choose appropriate materials, and optimize the engine's components for performance.

- Fabrication and Assembly:

  Objective: Manufacture the engine components and assemble them into a functional jet engine. Key Activities: Use machining and fabrication techniques to create compressor blades, combustion chambers, turbines, and nozzles. Assemble the engine according to the design specifications.

- Testing and Performance Evaluation:

  Objective: Test the jet engine to evaluate its performance, efficiency, and reliability.

  Key Activities: Conduct static tests to measure thrust, analyze fuel consumption rates, assess temperature and pressure variations, and evaluate overall engine stability.

- Safety and Reliability:

  Objective: Ensure the jet engine operates safely and reliably under various conditions.

  Key Activities: Implement safety features, conduct stress tests to identify potential weaknesses, and refine the design to enhance reliability and durability

1.2 Motivation

The development of the jet engine can be attributed to a combination of scientific curiosity, engineering innovation, and the drive for progress and efficiency in the field of transportation and aviation. Here are several motivating factors that have contributed to the creation and advancement of jet engines:

- Speed and Efficiency:

  Jet engines allow aircraft to travel at much higher speeds compared to traditional πston engines. This speed not only reduces travel time but also enhances efficiency, making air travel faster and more convenient.

- Global Connectivity:

  Jet engines have played a πvotal role in shrinking the world by making international travel accessible and efficient. They have connected people, cultures, and economies, fostering global understanding and collaboration.

- Military Applications:

  The military saw the strategic advantage of high-speed aircraft during times of conflict. Jet engines enabled the development of supersonic fighter jets and reconnaissance aircraft, enhancing national security capabilities.

- Scientific Curiosity:

  Scientists and engineers have always been motivated by the desire to explore the unknown. The challenge of harnessing the power of high-speed jet propulsion sparked curiosity and led to groundbreaking research and experimentation.

- Technological Innovation:

  Creating a jet engine required significant advances in materials, aerodynamics, and thermodynamics. The pursuit of these innovations not only drove the development of jet engines but also led to breakthroughs in other fields of science and engineering.

1.3 Purpose

Creating a jet engine as a college project can serve several valuable purposes, enhancing your learning experience and skillset in various ways:

- Hands-on Learning:
  Building a jet engine provides a practical, hands-on experience that supplements theoretical knowledge. It allows you to apply classroom concepts to real-world engineering challenges.

- Engineering Skills Development:
  Designing, assembling, and testing a jet engine can improve your skills in mechanical engineering, thermodynamics, aerodynamics, and materials science. It enhances your understanding of complex engineering principles.

- Teamwork and Collaboration:
  Collaborating with peers on a project of this scale teaches teamwork, communication, and project management skills. You'll learn how to divide tasks, share responsibilities, and work cohesively towards a common goal.

- Problem-Solving Abilities:
  Building a jet engine involves overcoming numerous technical challenges. You'll develop problem-solving skills by addressing issues related to design, fabrication, and performance optimization.

- Innovation and Creativity:
  Creating a jet engine encourages innovation and creative thinking. You might need to devise unique solutions to problems, encouraging you to think outside the box and explore unconventional approaches.

- Practical Application of Theory:

This project allows you to see theoretical concepts come to life. You'll gain a deeper understanding of the principles you've studied and their real-world applications.

- Networking Opportunities:

  Engaging in such a project can attract attention from professors, industry professionals, and potential employers. It might lead to networking opportunities and even job offers, especially if your project demonstrates exceptional skills and innovation.

- Preparation for Future Endeavors:

  If you plan to pursue a career in aerospace engineering, mechanical engineering, or related fields, this project will provide you with a strong foundation and valuable experience that can set you apart in job interviews and graduate school applications.

Chapter 2: Literature Survey

This chapter presents the information that was found during the survey conducted. Several research papers and books were used as references and the information pertaining to this project was found and is presented

Technology for Jet Engines: Case Study in Science and Technology Development (Paul D.Collopy)

Large jet engine research and development in the United States from the 1960s through the establishment of the Integrated High Performance Turbine Engine Technology Program in the late 1980s and the role of development in advancing fielded airbreathing propulsion capabilities are examined. The focus is on science and technology management principles employed during that period and how they impacted the process by which technologies were introduced into production aircraft.

Design & Construction of a single turbojet engine (Rikard Ilhl-Roos & Simon Fahlstrom)

The design was made by studying the work done by industry and researchers over the course of the history of jet engines. The methods were then discussed and chosen in a way that would simplify the design work as well as the construction of the engine. The goal was to create a self-sustaining combustion within the engine.

A Study Paper on Jet Engine (Himanshu Singh, Himanshu Somvanshi & Santosh Shinde)

This Paper deals with a brief article about a jet engine and its working explained along with its various types. A Jet Engine is a reaction Engine that discharges a fast moving jet which generates thrust by jet propulsion in accordance with Newton's Law of Motion. In common parlance, the term jet engine loosely refers to an internal combustion air breathing jet engine

(a duct engine). These tyπcally consist of an engine with a rotary (rotating) air compressor powered by a turbine (–Brayton Cycle‖), with the leftover power providing thrust via a propelling nozzle. A young German physicist, Hans von Ohain, worked for Ernst Heinkel, specializing in advanced engines, to develop the world's first jet plane, the experimental Heinkel. He 178. It first flew on August 27, 1939.

Ultrahigh-Temperature Materials for Jet Engines (Cambridge University Press)

This introductory article provides the background for the September 2003 issue of *MRS Bulletin* on Ultrahigh-Temperature Materials for Jet Engines. It covers the need for these materials, the history of their development, and current challenges driving continued research and development.

EARLY ASSESSMENT OF DEFECTS AND DAMAGE IN JET ENGINES (Joerg R. Seume, Rafael R. Adamczuk)

The defects under investigation include the malfunction of one burner, the increase the turbine blade tip clearance and burned trailing edges of the blades. The changes in the pattern resulting from the defects are compared to the density distribution of the undamaged initial state. It is shown that different exhaust jet patterns can be linked to the investigated hot gas path defects.

Design and Analysis of Centrifugal Impeller (Subash K , Muthukumar K)

In this research paper, the performance of the centrifugal pumps can be improved by alternating the geometric characteristics of impeller in the pumps. Centrifugal pump performance mainly depends upon blade exit angle. The vibration is mainly caused by blade exit angle. The rotating of the blade is based on the natural frequency and mode shapes. The impeller chosen for the study was a single- entry impeller with radial vanes.

Design and calculations of a combustion chamber for a turbojet engine (Shivam Sudarshan Vermaa Abhipsa, Dasbabub Parikshit, Baruahb Rohit, Dasb Anwesha, Dasbabub, Lovely Professional University Phagwara, Punjab)

Jet engine is an energy conversion device. It converts chemical energy into a rotatory mechanical energy. It works on the concept of Brayton Cycle. In this work, a combustion chamber has been designed for a turbocharger-based jet engine. Calculations have been done for the designing of a combustion chamber in the engine.

Chapter 3: Design Methodology

This chapter describes the meticulous process that incorporates actual dimensions and employs Computer-Aided Design (CAD) modeling to ensure accuracy and real-world applicability

3.1 Components

Impeller:

An impeller is a driven rotor utilized to increment the weight and stream of a liquid. It is the inverse of a turbine, which extricates vitality from, and decreases the weight of, a streaming liquid.

Strictly speaking, propellers are a sub-class of impellers where the stream both enters and takes off pivotally, but in numerous settings the term "impeller" is saved for non-propellor rotors where the stream enters pivotally and takes off radially, particularly when making suction in a pump or compressor. The makes a difference us to convey the fundamental oxygen required for the combustion process.

Material: Polylactic Acid (PLA)

PLA (Polylactic Acid) is a biodegradable, sustainable and food safe polymer made from organic sources. It is the most common used filament in FFF 3D printers for its ease of use and awide range of applications, especially those not mechanically or thermally demanding. Definitely a good starting point to learn about the 3D Printing manufacturing process.

Properties

- Detailed and glossy surface quality
- Good tensile strength
- Rigid, fragile behaviour
- Good UV resistance
- Withstand operating temperatures up to 50 °C.
- Odor-free, ideal for educational and office environments
- Compatible with PVA supports
- Low humidity resistance

Table 1 - Mechanical Properties of PLA

Mechanical properties		
	Typical value	Test method
MFR 210°C/2,16 kg	9,56 gr/10 min	ISO 1133
Tensile strength at yield	70 Mpa	ISO 527
Strain at yield	5 %	ISO 527
Strain at break	20 %	ISO 527
Tensile Modulus	3120 MPa	ISO 527
Impact strength-Charpy method 23°C	3.4 kJ/m²	ISO 179
Moisture absorption	1968 ppm	ISO 62

AutoCAD Modelling:

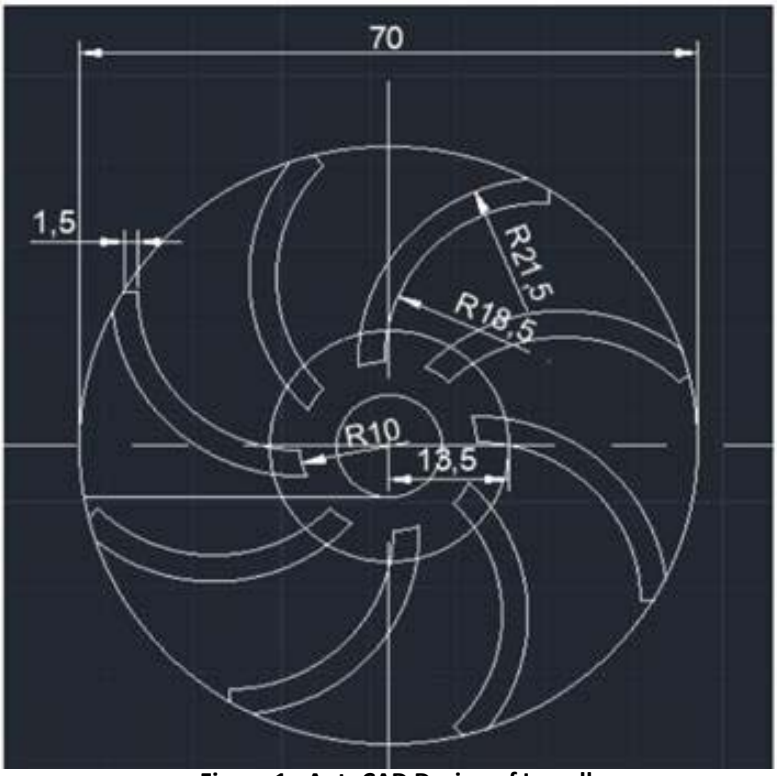


Figure 1 - AutoCAD Design of Impeller

3D Printed Impeller:



Figure 2 - 3D Printed Impeller

Table 2 - Natural frequency table used as reference

Mode	Frequency (Hz)					
	β = 20°	β = 21°	β = 22°	β = 23°	β = 24°	β = 25°
1	755.29	754.63	759.27	761.77	766.89	769.07
2	756.2	755.73	760.0	762.74	768.8	770.62
3	916.59	914.88	915.99	916.22	917.88	918.01
4	1220.3	1217.8	1220.1	1221.4	1225.7	1227.4
5	1220.7	1218.3	1220.6	1222.0	1226.2	1227.9
6	1959.8	1979.8	1986.3	1990.9	1995.7	1998.6

Stator:

A Stator is a mechanical gadget that increments the weight of a gas by lessening its volume. An discuss stator is a particular sort of gas compressor. Stators are comparable to pumps: both increment the weight on a liquid and both can transport the liquid through a pipe. The primary refinement is that the center of a compressor is to alter the thickness or volume of the liquid, which is for the most part as it were achievable on gasses. Gasses are compressible, whereas fluids are moderately incompressible, so compressors are once in a while utilized for fluids. The primary activity of the stator in our fly demonstrate is to decrease the tall speed oxygen into a compressed shape and supply it assist in the combustion chamber for burning the fuel.

The stator works as inactive compressor for the discuss which is continuously under tall supply since of the turning impeller. The little opening in Stator makes a difference to supply constrained sum of discuss which gets pressurized since of nonstop weight getting construct up since of impeller as well.

This makes an indicated sum of mass stream all through the combustion chamber for wanted effectiveness and superior butane discuss mixture.

The stator moreover has the required openings for holding the engine which is utilized for turning the impeller.

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Tensile Modulus	3120 MPa	ISO 527
Impact strength-Charpy method 23°C	3.4 kJ/m²	ISO 179
Moisture absorption	1968 ppm	ISO 62

Solidworks Model of 1st Iteration of Stator:

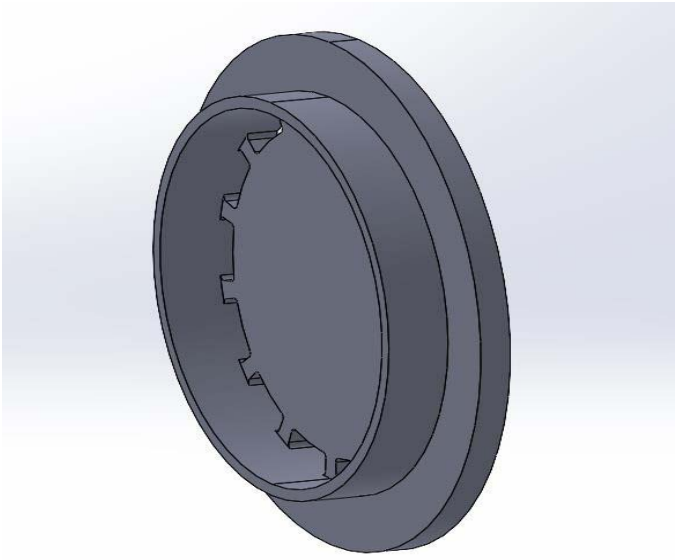


Figure 3 - Stator Design without motor housing

Solidworks Model of 2nd Iteration of Stator:

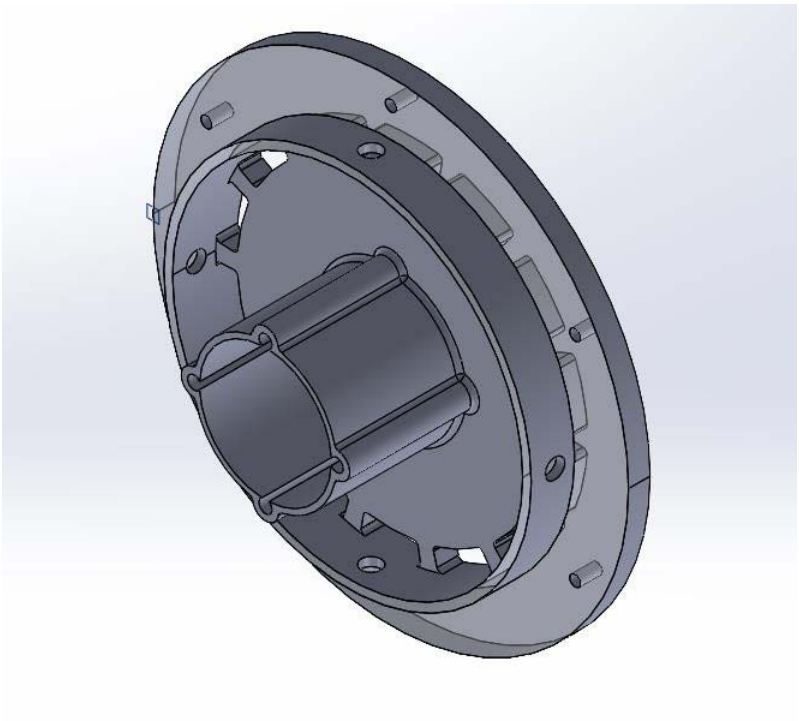


Figure 4 - Stator Design with Motor Housing

3D Printed Model:



Figure 5 - 3D Printed Stator

Calculations:

Diameter of inlet = 26mm, r = 13mm

Area of inlet = $A = \pi r^2$

= 3.142 x 13 x 13

= 530.998 mm^2

Assuming the area required for the stator of a jet engine is twice the area of inlet

Area of Stator = 530.998 x 2

= 1061.1996 mm^2

Designing slots of air:

Length of slot = 15mm

Breadth of slot = 5mm

Area of each slot = l x b

= 15 x 5 = 75 mm^2

Number of slots required = 1061.1996 / 75

= 14.14 (approx. 14)

= 14 slots

Motor:

As their name implies, brushless DC motors do not use brushes. With brushed motors, the brushes deliver current through the commutator into the coils on the rotor. So how does a brushless motor pass current tothe rotor coils? It doesn't—because the coils are not located on the rotor. Instead, the rotor is a permanent magnet; the coils do not rotate, but are instead fixed in place on the stator because the coils do not move, there is no need for brushes and a commutator.

With the brushed motor, rotation is achieved by controlling the magnetic fields generated by the coils on the rotor, while the magnetic field generated by the stationary magnets remains fixed. To change the rotation speed, you change the voltage for the coils. With a BLDC motor, it is the permanent magnet that rotates; rotation is achieved by changing the direction of the magnetic fields generated by the surrounding stationary coils. To control the rotation, you adjust the magnitude and direction of the current into these coils.

We are using a 1400 RPM motor to rotate our impeller.



Figure 6 - BLDC Motor



Figure 7 - BLDC Motor with attachment

Specifications:

Speed	1400 RPM
Voltage	12 Volts
Material	Metal, Steel with Aluminium attachment
Item Weight	150 Grams
Max. Efficiency Current	6-12 A

Speed Controller:

The speed controller is used to control the rpm speed varying between the range of 1000-1400 RPM and is used to control the oxygen supply through the stator. This helps to control the combustion process and the oxygen supply can be limited through the speed controller.



Figure 8 - Speed Controller

Model Name: CCPM Servo Consistency Master

Model Specifications:

- Operating Voltage: 4.8 – 6 Volts
- Operating Temperature: -40 – 800C
- Signal Output: 1.5ms±0.5ms
- Output Current: >15 mA (5 mA per berg)
- 3X3 berg to connect 3 servo motors simultaneously
- modes of operations
- A knob to rotate servo manually
- Tiny box with dimensions in millimeters

Table 4 - Pin Configuration

Pin Name	Function
INS	Signal Input pin
IN+	Power Supply pin
IN-	Ground pin
OUT_S	3×3 connector to connect three signal outputs to three servo motors
OUT+	Servo motor power pin
OUT-	Servo motor ground pin

ESC:

Standard BLDC 30 amp ESC Electronic Speed Controller with Connector is specifically made for quadcopters and multi-rotors. It provides faster and better motor speed control giving better flight performance compared to other available ESCs.

STANDARD 30 amp ESC (Electronic Speed Controller) can drive motors that consume up to 30A current. It works on 2-3S LiPo batteries. The onboard BEC provides regulated 5V (2A max draw) to power the flight controller and other onboard modules.
This is useful to control our brushless motors with a 2-3S LiPo (make sure the motor doesn't draw more than 30A).

This particular ESC is recommended with an A2212 brushless motor (1000kv, 1400kv, 2200kv) for more motor,



Figure 9 - ESC Module

Specifications:

Table 5 - ESC Specifications

Model	Standard 30A BLDC ESC
Burst Current (A)	40
Constant Current (A)	30
BEC	Yes (5V/3A)
Suitable Lipo Batteries	2 ~ 3S Lipo and 4 ~ 10s Ni-Mh/Ni-CD
Color:	Random
Application	BLDC Motors, Multirotors, Rc Planes etc.
Length (mm):	45
Width (mm):	24
Height (mm):	9
Weight (g):	23
Shipping Weight	0.027 kg
Shipping Dimensions	10 × 10 × 5 cm

Combustion Chamber:

A combustion chamber is part of an internal combustion engine in which the fuel/air mix is burned. For steam engines, the term has also been used for an extension of the firebox which is used to allow a more complete combustion process. In an internal combustion engine, the pressure caused by the burning air/fuel mixture applies direct force to part of the engine (e.g. for a piston engine, the force is applied to the top of the piston), which converts the gas pressure into mechanical energy (often in the form of a rotating output shaft). This contrasts an external combustion engine, where the combustion takes place in a separate part of the engine to where the gas pressure is converted into mechanical energy.

In this combustion chamber holes in varying sizes are drilled so as to control the air supply for the combustion. The holes allow the oxygen to enter in high pressure but slow speed which is the crucial part of combustion process. The combustion chamber is made of stainless steel to handle the high temperature caused during the combustion process.

The flame tube is designed to mix a small percentage of the air that enters the combustion chamber with the fuel and control the burning process so that the flame never actually touches the walls of the chamber or extends into the turbine area.

Material: SS 304

Material Properties:

Table 6 - Mechanical Properties of SS 304

Type 304 Physical Properties	Metric
Density	7.93 g/cm3
Melting point	1400-1450 °C
Specific heat capacity	500 J/kg·K at 20 °C
Electrical resistivity	0.72 μΩ·m at 20 °C

Conductivity, %, IACS	2.5
Magnetic permeability	1.02 (Approximate)
Elastic modulus (Modulus of Elasticity)	193 GPa
Thermal diffusivity	3.84 mm2/s
Thermal conductivity	16.2 W/m·K at 100 °C
	21.5 W/m·K at 500 °C
Coefficient of thermal expansion (CTE)	17.2 10^{-6}/K at 0-100 °C
	17.8 10^{-6}/K at 0-315 °C
	18.4 10^{-6}/K at 0-538 °C

Solidworks Model of the Flame Tube:



Figure 10 - CAD Model of Flame Tube



Figure 11 - Flame Tube

Calculations:

Calculation for flame tube = 2 x inlet diameter

= 2 x 26 mm

=52mm (approx. 50mm)

Calculations for hole zone in flame tube:

Primary Hole zone:

Total area of holes in primary hole zone = 30%

Area of inlet = π x r2 = 3.14 x 132 = 530.998 mm^2

Total area of holes in primary zone = 30/100 x 530.998

= 159.29 mm^2

Since there are 17 holes in primary zone = 159.29/17

Area of one hole = 9.37 mm^2

Therefore, R2 = 9.37/π = 2.9823

R = 1.726 (approx. = 1.75mm)

Therefore, Diameter of hole = 3.5mm

Secondary Hole zone:

Total Area of holes in secondary zone = 20%

Area of inlet = 530.998 mm2

Total Area of holes in secondary zone = 20/100 x 530.998 = 106. 1996mm2

Since there are 6 holes in secondary zone = 106. 1996/6

= 17.6993mm2

Area of 1 hole = 17.6993mm2

Therefore, R2 = 5.6333

R = 2.37 mm.

D = 4.74 mm (approx. 5mm)

Tertiary Hole Area of Zone:

Total area of holes in Tertiary hole zone = 50%

Area of inlet = 530.998 mm2

Total area of holes in Tertiary hole zone = 50/100 x 530.998

= 265.499 mm2

Since there are 5 holes in tertiary zone = 265.499/5 = 53.0998mm2

Area of 1 hole = 53.0998mm2

Therefore, R2 = 53.0998/π = 16.9

R = 4.110 mm

D = 8.22 mm (approx. 8mm)

Net Weight:

We use this formula to get the linear density, μ, of a pipe:

$\mu = \pi \times [(Do^2)/4 - (Di^2)/4] \times \rho$

$= 3.142 \times [(51^2/4) - (49.5^2/4)] \times 8.03$

$= 958.74$

Where, Do = Outer Diameter

Di = Inner Diameter

ρ = Density of the pipe material

We can then take the weight of the pipe by multiplying the linear density by the length of the pipe, as shown in this formula:

$W = \mu \times L$

$= 958.74 \times 80$

$= 76699.904$ mgs

$= 76.69$ gms

Where, L= Length of the pipe

W =Weight of the pipe

Housing/Casing:

The housing, or casing, of a jet engine refers to the external structure that contains and supports the various components of the engine. It serves several important functions:

- Containment: The housing encloses the internal components of the jet engine, such as the compressor, combustion chamber, turbine, and exhaust nozzle, protecting them from external elements and providing structural integrity.

- Aerodynamics: The housing is designed to optimize the airflow through the engine, ensuring efficient operation and minimizing drag.

- Mounting: It provides attachment points for mounting the engine to the aircraft's structure.

- Cooling: The housing may incorporate features for airflow management and heat dissipation to maintain optimal operating temperatures for the engine components.

- Noise Reduction: Modern jet engine housings often include features for noise reduction to mitigate the impact of engine noise on both passengers and the environment.

The design and materials used in the housing depend on various factors, including the type of aircraft, engine configuration, operating conditions, and manufacturing considerations. Typically, jet engine housings are made from high-strength materials such as titanium alloys, aluminium alloys, or composite materials to withstand the high temperatures, pressures, and forces experienced during operation.

Material: SS 304

Material Properties:

Table 7 - Mechanical Properties of SS 304

Type 304 Physical Properties	Metric
Density	7.93 g/cm3
Melting point	1400-1450 °C
Specific heat capacity	500 J/kg·K at 20 °C
Electrical resistivity	0.72 μΩ·m at 20 °C
Conductivity, %, IACS	2.5
Magnetic permeability	1.02 (Approximate)
Elastic modulus (Modulus of Elasticity)	193 GPa
Thermal diffusivity	3.84 mm2/s
Thermal conductivity	16.2 W/m·K at 100 °C
	21.5 W/m·K at 500 °C
Coefficient of thermal expansion (CTE)	17.2 10^{-6}/K at 0-100 °C
	17.8 10^{-6}/K at 0-315 °C
	18.4 10^{-6}/K at 0-538 °C

Welding of Flame tube with Outer Casing:



Figure 12 - Welded Housing

Calculations:

Clearance of housing/Casing:

Diameter of flame tube + 15mm both sides

So, diameter of case is approx. 80mm

Our Casing Diameter = 89mm which is greater than 80mm, so it is acceptable.

Net Weight:

We use this formula to get the linear density, μ, of a pipe:

$\mu = \pi \times [(Do^2)/4 - (Di^2)/4] \times \rho$

$= 3.142 \text{ x } [(89^2/4) - (85^2/4)] \text{ x } 8.03$

$= 3.142 \text{ x } [1980.25\text{-}1806.25] \text{ x } 8.03$

$= 4390.065$

Where, Do = Outer Diameter

Di = Inner Diameter

ρ = Density of the pipe material

We can then take the weight of the pipe by multiplying the linear density by the length of the pipe, as shown in this formula:

$W = \mu \times L$

$= 4390.065 \text{ x } 150$

$= 658509.786 \text{ mgs}$

$= 658.50 \text{ gms}$

Where, L= Length of the pipe

W =Weight of the pipe

Impeller Casing:

Impeller casing plays one of the crucial roles in Jet engine as it an important of the compression of the oxygen sucked by the impeller. The impeller has a certain opening specially designed to intake certain amount of air and pass it further to the impeller which the pressurizes it through the stator.

The impeller casing has an opening 26mm Diameter. This is an assumed value with respect to the hub of the impeller and it is assumed for ease in the designing process of the jet engine.

3D printed Model:



Figure 13 - 3D Printed Impeller Casing

Nozzle:

A nozzle is a relatively simple device, just a specially shaped tube through which hot gases flow. However, the mathematics which describe the operation of the nozzle takes some careful thought. As shown above, nozzles come in a variety of shapes and sizes depending on the mission of the aircraft. Simple turbojets, and turboprops, often have a fixed geometry convergent nozzle.

The design of the nozzle is carefully engineered to ensure efficient exhaust gas expansion and minimize losses. The nozzle is part of the exhaust system and is responsible for controlling the expansion of exhaust gases before they exit the engine. By adjusting the size and shape of the nozzle, engineers can optimize the exhaust flow velocity and pressure to maximize engine performance and efficiency.

Like other components of a jet engine, the nozzle requires regular inspection and maintenance to ensure proper functionality and safety. Inspections may include checks for wear, erosion, or damage due to high-temperature exposure, and maintenance procedures may involve repair or replacement as needed.

Material: SS 304

Nozzle Type: Convergent



Figure 14 - Nozzle

Calculations:

Desired thrust (F): 1 kg (approximately 9.81 N due to the acceleration due to gravity)

Mass flow rate of air ($\dot{m}$air): 0.099 kg/s

Air-fuel ratio: 15.42:1

Specific impulse (Isp) for butane: 70 s (seconds)

Calculate the mass flow rate of fuel ($\dot{m}$fuel):

Using the air-fuel ratio:

$\dot{m}$fuel = ($\dot{m}$air)/ (Air-fuel ratio)

$\dot{m}$fuel = (0.099 kg/s) / (15.42)

$\dot{m}$fuel ≈ 0.00642 kg/s

Calculate the total mass flow rate ($\dot{m}$total):

$\dot{m}$total = $\dot{m}$air + $\dot{m}$fuel

$\dot{m}$total = 0.099 kg/s + 0.00642 kg/s

$\dot{m}$total ≈ 0.10542 kg/s

Calculate the exhaust velocity (Ve):

Using the specific impulse:

$Ve = Isp \times g$

Ve = 70 s × 9.81 m/s^2

$Ve \approx 686.7$ m/s

Calculate the throat area (A_t):

$A_t = \dot{m}\text{total}/\rho \times a$

The density (ρ) and sonic velocity (a) depend on the specific conditions and need to be determined based on the temperature and pressure of the exhaust gases. However, for simplification, let's assume standard conditions and use the values for air ($\rho \approx 1.225$ kg/m^3) and sonic velocity ($a \approx 343$ m/s).

$A_t = (0.10542 \text{ kg/s}) / (1.225 \text{ kg/m}^3 \times 343 \text{ m/s})$

$A_t \approx 0.000254 \text{ m}^2 \text{ x } 10^6$

$A_t \approx 254 \text{ mm}^2$

If we assume a convergent nozzle, the exit area (Ae) will be equal to the throat area (A_t). This is because a convergent nozzle has a constant area from the throat to the exit.

Given that $Ae = A_t = 254\, mm^2$, the exit diameter (De) can be calculated using the formula for the area of a circle:

$Ae = \pi \times (De/2)^2$

Rearranging the formula to solve for De:

$De = \sqrt{(4 \times Ae/\pi)}$

$De = \sqrt{(4 \times 254/3.142)}$

$De \approx 18.02\, mm$

So, for a convergent nozzle, the exit diameter (De) would be approximately 18.02 mm.

Fuel Line:

The fuel line is one of the essential components for the supply of flammable liquid necessary for the combustion. The fuel is directly placed inside the combustion chamber for it to react with the oxygen for combustion and thrust can be generated. It is made up of stainless steel for maximum heat resistance.

Material: SS 304

Melting Point: 1400 – 1450 °C

Pipe Diameter = 8 mm

Hole Diameter = 2 mm



Figure 15 - Fuel Line with 2mm holes

Fuel - Butane:

Butane is primarily used as a fuel for portable stoves, camping stoves, lighters, and torches due to its clean-burning properties. It is also used as a propellant in aerosol sprays and as a refrigerant in some applications.

Properties:

Chemical formula: C_4H_{10}

Molar mass: 58.124g·mol−1

Appearance: Colorless gas

Self-Ignition Temperature: 420 (°C) & 788 (°F)

Density: 2.48 kg/m3 (at 15 °C (59 °F))

Melting point: −140 to −134 °C; −220 to −209 °F; 133 to 139 K

Boiling point: −1 to 1 °C; 30 to 34 °F; 272 to 274 K



Figure 16 – Butane Gas

Spark Plug:

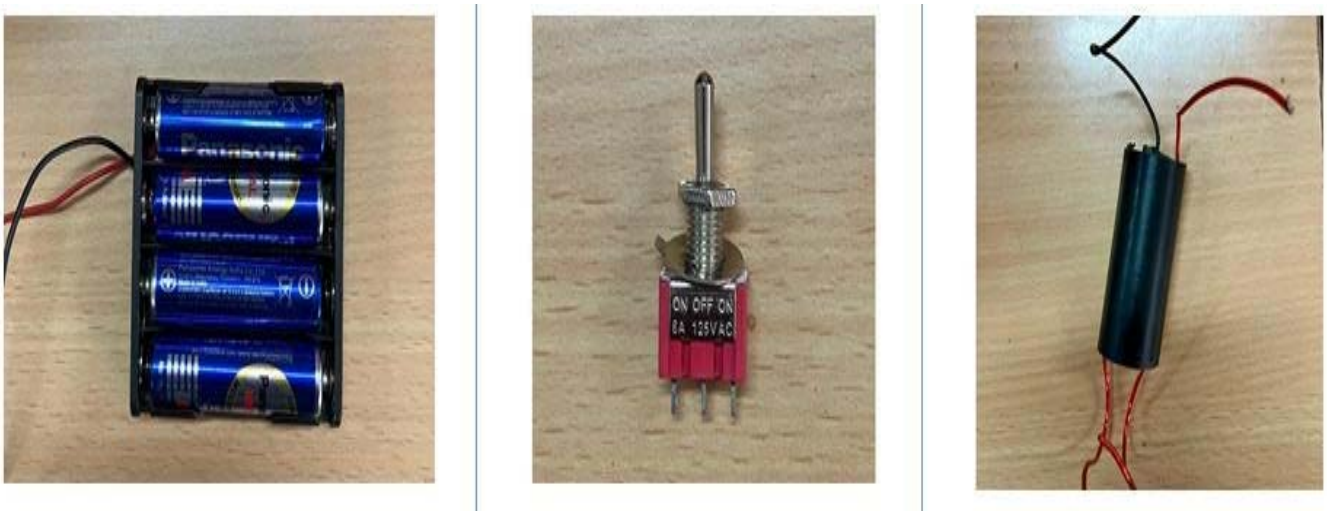


Figure 17 – Spark Plug Components

The figure on the left hand side is the battery pack used to supply voltage into the generator.

The figure in the middle is the ON/OFF switch for the spark plug.

The figure on the right hand side is the high voltage generator used for generating the spark.

Cell Used: 1.2 V Pencil Cell

Battery:

We are using rechargeable 12V LiPo Battery as our power unit to rotate the impeller.

The ORANGE 1300mAh 3S 40C (11.1 V) Lithium Polymer Battery Pack (Li-PO) are known for performance, reliability, and price. It's no surprise to us that Orange Lithium polymer packs are the go-to pack for those in the know. The Orange batteries deliver the full rated capacity at a price everyone can afford.

The ORANGE 1300mAh 3S 40C (11.1 V) Lithium Polymer Battery Pack (Li-PO) are equipped with heavy-duty discharge leads to minimize resistance and sustain high current loads. Orange batteries stand up to the punishing extremes of aerobatic flight and RC vehicles.

Features :

1. Product Type:Lithium Polymer Battery Pack
2. The Orange LiPo battery has matched resistance.
3. Good Temperature Control.
4. Minimum weight in Class



Figure 18 – 12V LiPo Battery

Model Specifications:

Table 8 - Battery Specifications

Model No.	ORANGE 1300mAh 3S-40C
Output Voltage (VDC)	11.1
Charge Rate (C)	1 ~ 3
Discharge Plug	XT-60
Balance Plug	JST-XH
Length (mm):	74
Width (mm):	32
Height (mm):	22
Max. Burst Discharge (C)	50 C (10 Sec)
Max. Charge Rate	5 C
Max. Continuous Discharge	40C
Shipping Weight	0.215 kg
Shipping Dimensions	8 × 4 × 3 cm

Thickness Calculation with Barlow's Formula:

P = 2St/D (P & S is in bar, t & D is in mm)

1.5 x 89 = (2 x 110.316 x t)

t = 0.06 mm (Theoretical thickness)

Where P= Internal Pressure (in bar)
S= Allowable stress of Material
T= Wall thickness
D= Outside Diameter

3.2 CAD Modelling

AutoCAD:

The following AutoCAD file shows the complete 2D layout of the jet engine model. This includes components like Impeller casing, Impeller, Stator, Combustion Chamber and Reducer**.**

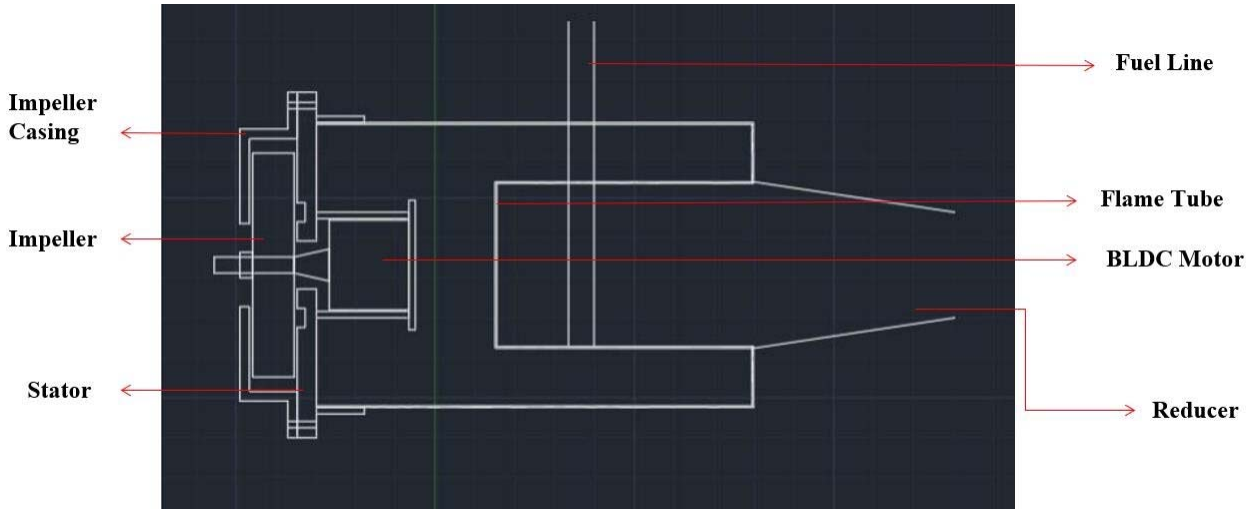


Figure 19 - Cross sectional view

Solidworks:

Side view:

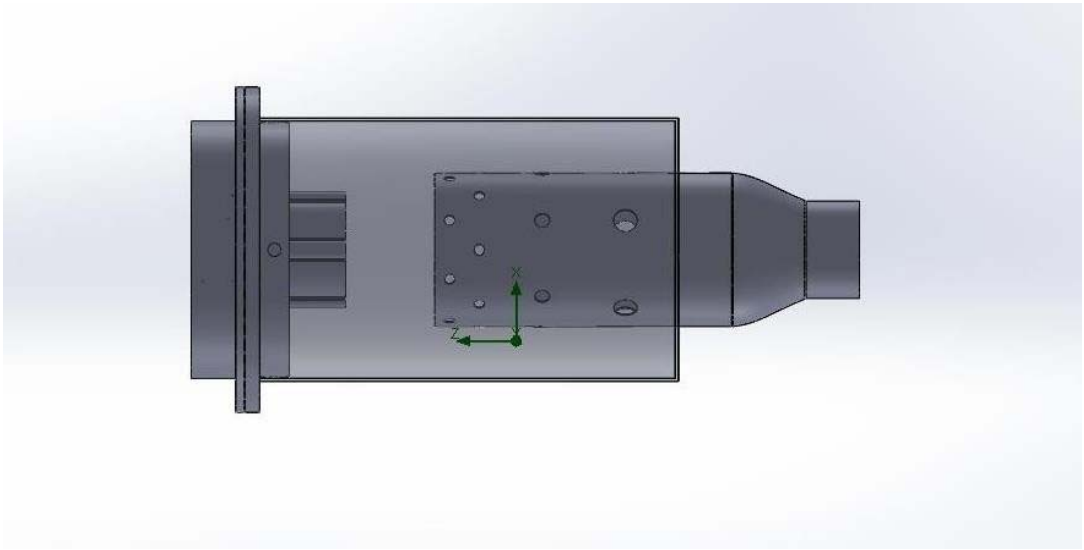


Figure 20 - CAD Design of Jet Engine Model

Isometric view:

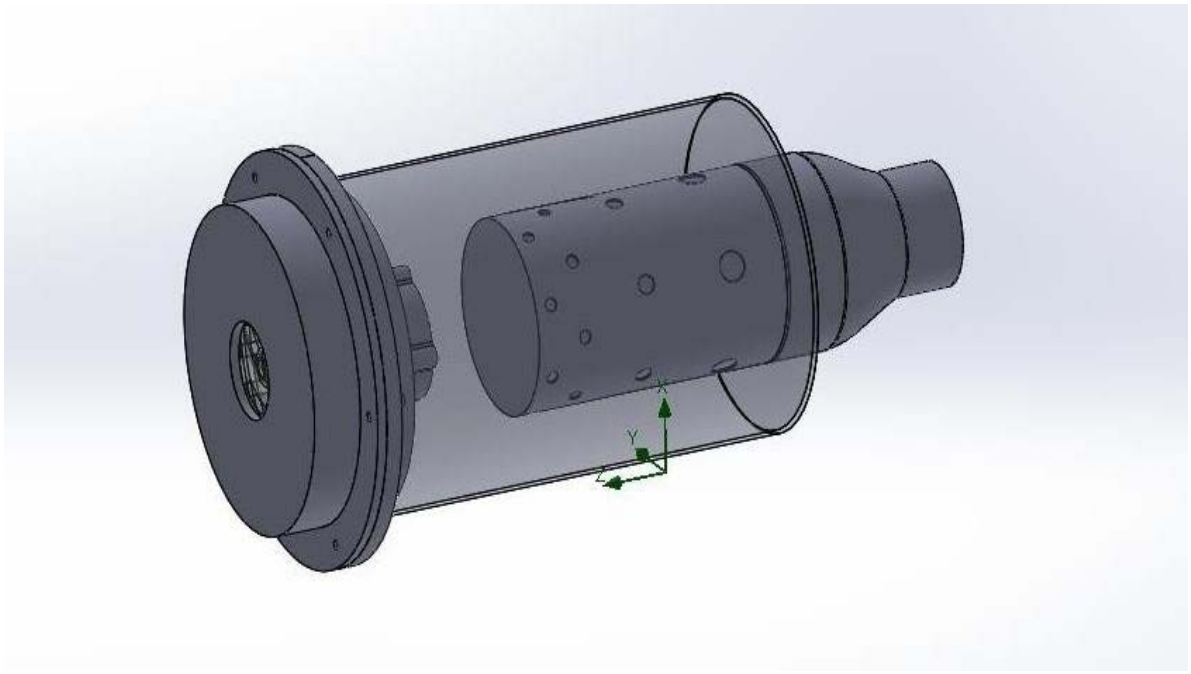


Figure 21 - Isometric View of Model assembly

Exploded View:

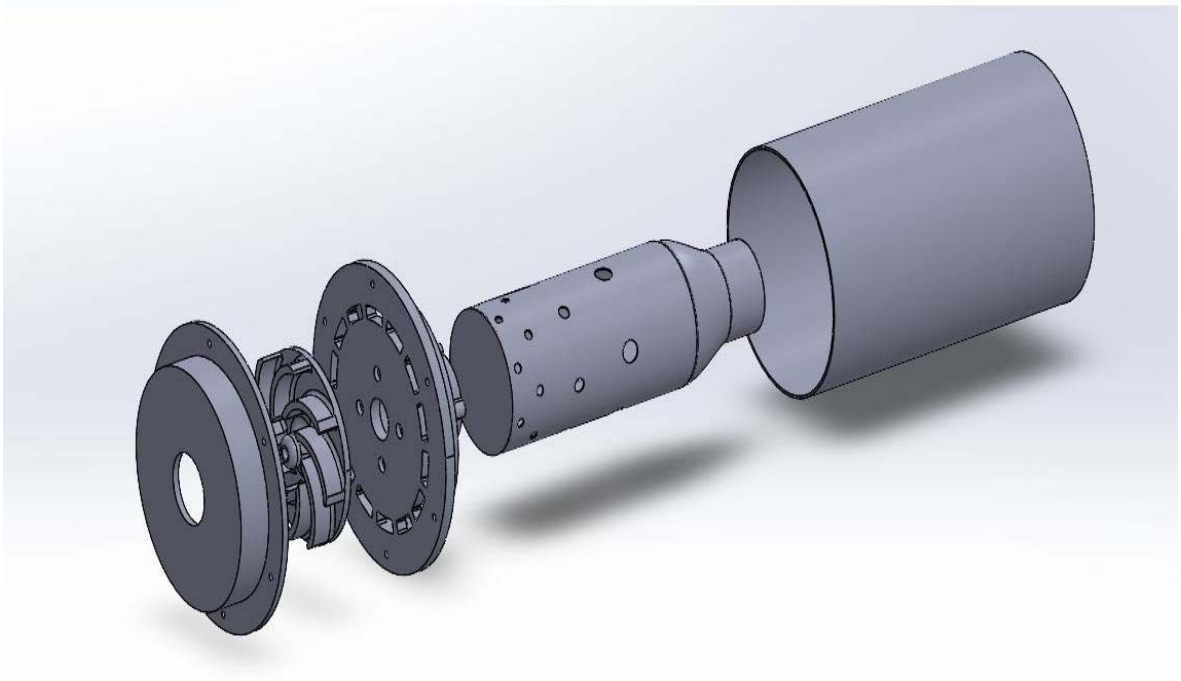


Figure 22 - Exploded View of Model Assembly

3.3 Flow Simulation

Solidworks Flow Simulation:

Flow analysis will help us to understand the air flow behavior with different important parameters such as Mass flow rate, Inlet outlet pressures along with inlet and exit velocity.

These parameters will help us to understand that will our system will be able to generate the required amount of thrust or not. The simulation shows how the air flow happens throughout the Jet engine and combustion chamber

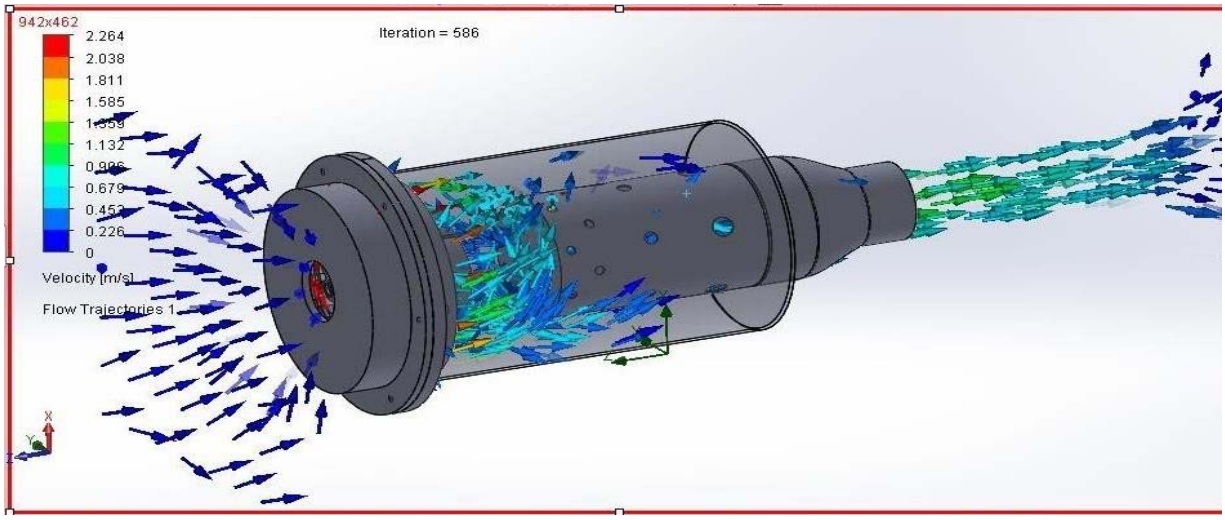


Figure 23 - Flow Simulation of AIr

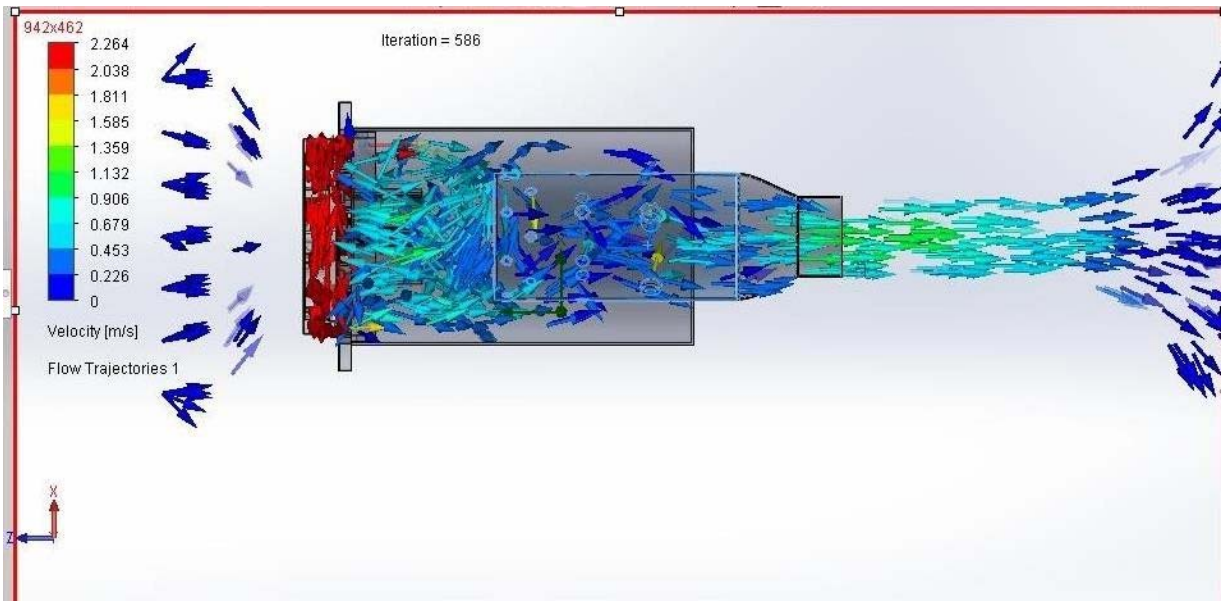


Figure 24 – Velocity Trajectory

Simulation Results:

Table 9 - Simulation Results

HYBRID JET ENGINE - 13mm Radius

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG inlet Static Pressure 1	[MPa]	0.101278331	0.1012785278	0.1012783206	0.1012787747	100	Yes	0.000000105	0.000252242
GG Average Static Pressure 2	[MPa]	0.101324278	0.1013242819	0.1013242774	0.1013242862	100	Yes	0.000000000	0.000000511
GG outlet Static Pressure 3	[MPa]	0.102449377	0.1023745273	0.1022997014	0.1024493779	100	Yes	0.000149669	0.000265576
GG Minimum Total Pressure 4	[MPa]	0.101281290	0.1012814278	0.1012812662	0.1012815899	100	Yes	0.000000015	0.000252247
GG Average Total Pressure 5	[MPa]	0.101324306	0.1013243095	0.1013243055	0.1013243143	100	Yes	0.000000000	0.000000511
GG Mass Flow Rate 7	[kg/s]	0.099342550	0.09765870489	0.09559997671	0.09949082113	100	Yes	0.003890844	0.004539697
GG Minimum Velocity 8	[mm/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 9	[mm/s]	46.44602029	44.33444627	42.17192497	46.44602029	8.8	Yes	4.274095325	0.379597756
GG Maximum Velocity 10	[mm/s]	12357.42275	12323.70446	12290.0823	12359.42476	100	Yes	2.494235347	604.1610477
GG Minimum Velocity (X) 11	[mm/s]	-2558.047789	-2555.94881	-2725.084066	-2538.748743	100	Yes	12.29691458	743.8637363
GG Average Velocity (X) 12	[mm/s]	16.39047755	15.42472095	14.44308968	16.39047755	7.1	Yes	1.947387872	0.138474329
GG Maximum Velocity (X) 13	[mm/s]	6825.263039	6830.434834	6662.058236	6854.479003	100	Yes	15.82496819	477.7974248
GG Minimum Velocity (Y) 14	[mm/s]	-10337.43762	-10333.93226	-10339.71158	-10324.33804	100	Yes	2.08267582	585.1421466
GG Average Velocity (Y) 15	[mm/s]	-0.046837020	-0.04730444915	-0.04877101267	-0.0458785024	35.1	Yes	0.002892510	0.001015341
GG Maximum Velocity (Y) 16	[mm/s]	9961.575222	9839.409841	9667.462899	10053.19804	100	Yes	7.19353793	467.6227152
GG Minimum Velocity (Z) 17	[mm/s]	-10251.40937	-10244.39781	-10251.97866	-10236.59431	100	Yes	4.03684586	261.3290959
GG Average Velocity (Z) 18	[mm/s]	0.050602484	0.04737886167	0.04353616279	0.05110321175	17.8	Yes	0.006573290	0.001170700
GG Maximum Velocity (Z) 19	[mm/s]	9857.306218	9856.124711	9853.93919	9858.162052	100	Yes	0.944617909	380.1508069

Output:

In this flow simulation, we get the necessary values of pressure, velocity, mass flow rate in order to calculate the thrust it can generate.

Chapter 4: Fabrication and Implementation

This chapter describes the various processes carried out and the various machine used in order to carry out the fabrication and manufacturing of the jet engine model without any setbacks.

4.1 Machining Processes:

Drilling:

Holes in the flame tube and fuel line are drilled with the help of vertical drill. Drilling Machine is the simplest, moderate, and most accurate machine tool used in almost all the production shops and tool rooms. Drilling is basically a single-purpose machine tool as its main purpose is to make holes in the workpiece.

Drill Bits Used: 2 mm, 3.5 mm, 5 mm, 8 mm

An AutoCAD layout was set for the drilling of holes in the flame tube. The layout was printed on 1:1 scale and then pasted on the pipe for drilling. The use of centre punch was done to first mark all the holes and avoid the tool from spinning out.

For 8mm holes, first 5mm holes are made and then 8mm is bit is used in the second pass. This helps in increasing the accuracy of the holes. Directly drilling 8mm holes is not advised in Stainless steel as it may lead to off-centering of the hole and damaging the drill bit.



Figure 25 – Drilling Process

Polishing:

These are of all the parts which will undergo Welding and other post processes. This includes outer Casing, Flame Tube, Reducer, Casing Ring and Fuel Line.



Figure 26 - Components of Jet Engine

After the finalization of each separate part, the parts are first polished to remove all the burr and smoothen the surface of the parts. It also removes rust from the surface of the metal.



Figure 27 – Polishing and Deburring

Machine used for Polishing Process: Die Grinder

A die grinder is a small, lightweight handheld grinder that's usually operated with one hand. Die grinders are used for grinding, sanding, and polishing. They are usually used for fine detailing and light work. Air-powered die grinders are used for heavier work, and electric die grinders are used for the lightest work. Die grinders abrade material using an abrasive disc or other attachment on the spindle.

Wheels used for the process: Wire brush wheel and Buffing wheel

Final Assembly:



Figure 28 – Welding Setup

After all the burr is removed from the parts, aligning of the parts is done to check whether there is some touch up work or is it okay to start the final assembly welding process.

A spacer is put in between the casing and the flame tube which helps in the assembly by keeping the flame tube in the centre and restricting any movement while welding is carried out.



Figure 29 – Spacer between Flame tube & Casing

Welding of the Flame Tube with the outer Casing and Casing Ring:

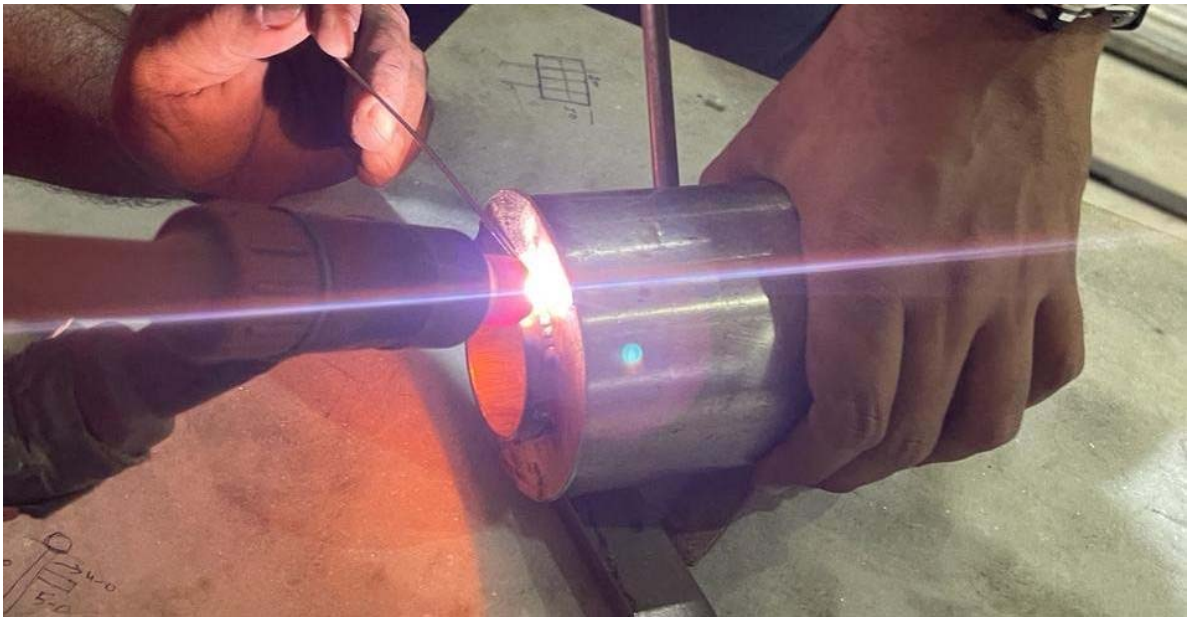


Figure 30 – Welding of Flame tube with Outer Casing

To hold the flame tube with outer casing, one thin Stainless-steel ring is fitted in-between them. The dimensions of the ring match with the outer diameter of both the flame tube and the casing. The fuel is always in position during this process.

We used MIG Welding to proceed with our fabrication.

Welding Wire used: ER308L

Wire Specifications:

- Yield Strength: 57,000 psi
- Tensile Strength: 85,000 psi
- Elongation: 40%
- Brinell Hardness: 160 HB

- Reduction of Area: 56%



Figure 31 – Final Assembly After Welding

After tack weld and further checks, the casing ring and the flame tube are cleared for welding. We encountered slight deformation in the ring due to the heat caused by welding. The spacer and the fuel line is removed after the welding is completed and the welding is tested for leaks by adding water and watching out for leaks via seepage.

Welding of Nozzle with Flame tube:

The final assembly of the parts of jet engine along with the nozzle reducer.



Figure 32 – Final Assembly of Nozzle with Flame tube

After the complete welding and testing, the fuel line is welded in the end.

Assembly of Impeller with Stator:

The motor is mounted into the motor casing on the back side of the staor. Then the impeller is mounted onto the shaft of motor, tightned with nut. Two washers were placed between the impeller and stator for purpose of having no contact between the stator surface and impeller blades.

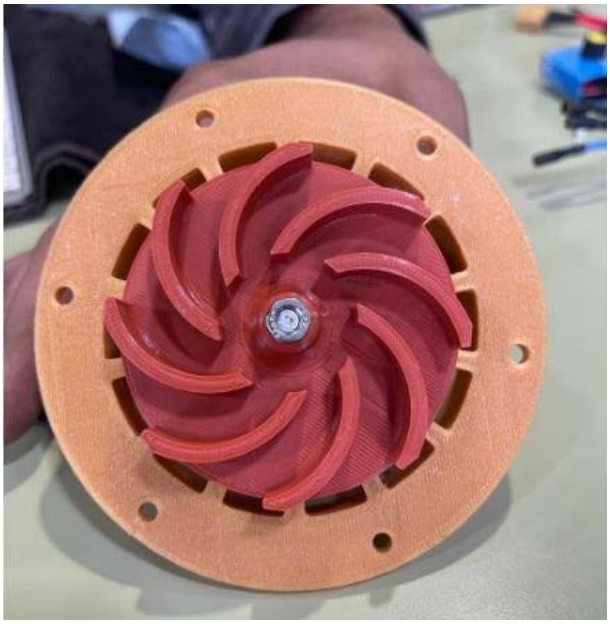


Figure 33 – Final Assembly of Impeller with Stator

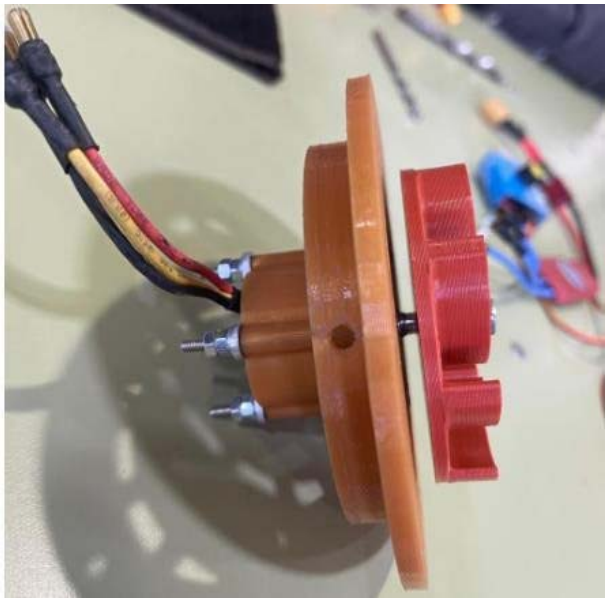


Figure 34 – Final Assembly of Impeller with Stator (Side View)

Complete Assembly of Jet Engine Model:

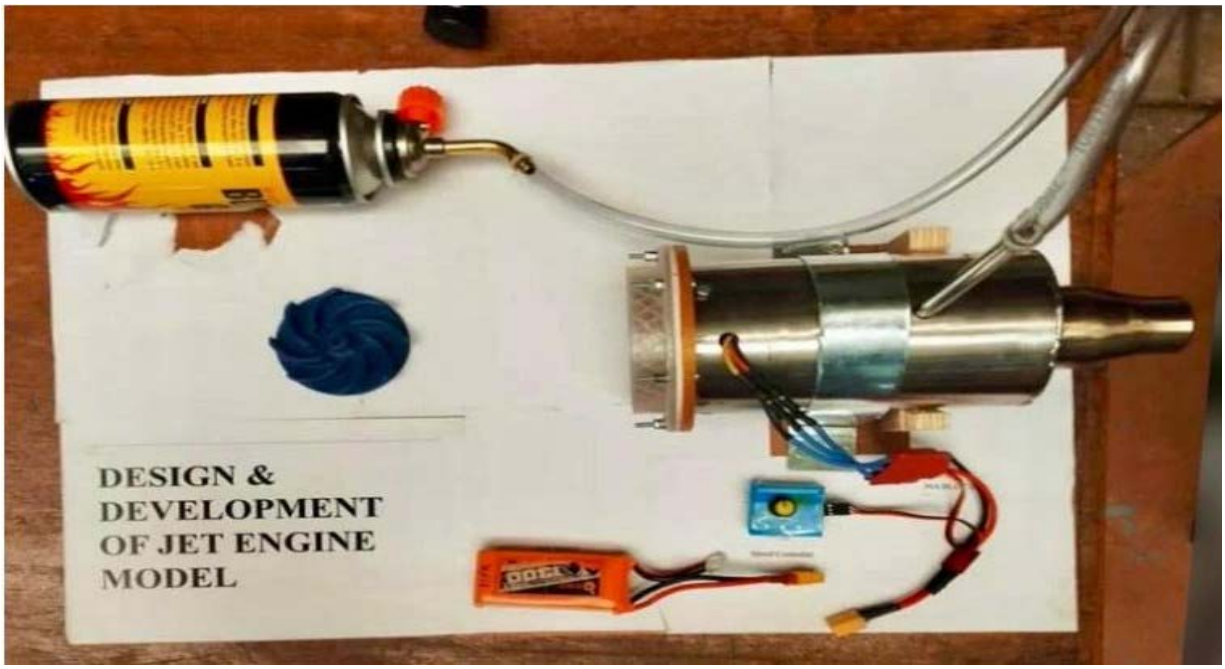


Figure 35 – Jet Engine Model

Total Weight of the Jet Engine Model after Assembly:

Our Model weighs a total of 1,271 grams or 1.2kg



Figure 36 – Model Weight without Electronics

Thrust Generated of Jet Engine Model:



Figure 37 – Thrust Generated

Working Model:

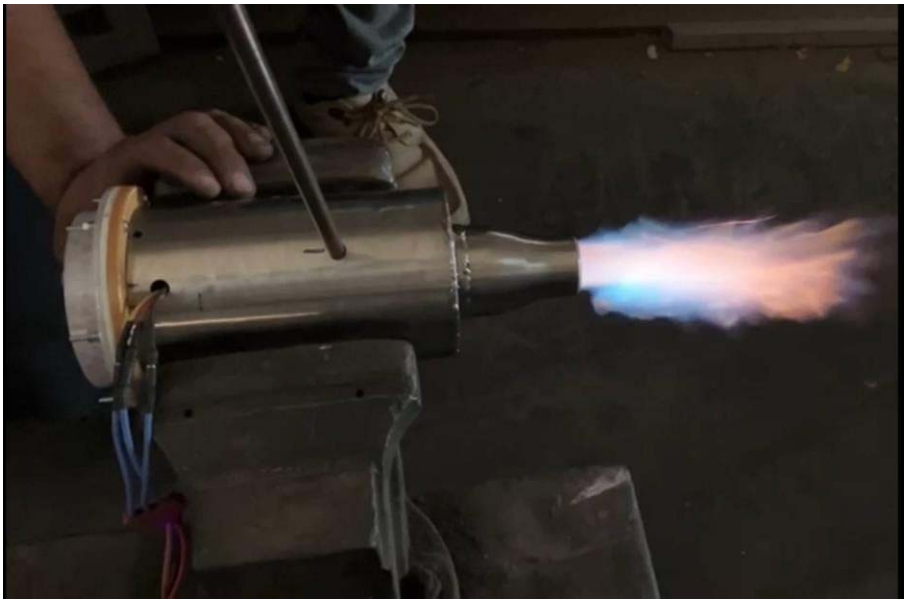


Figure 38 – Working Model

Bill of Materials (B.O.M):

Item No.	Description	Quantity	Unit Cost	Total Cost
1	3D Printed Impeller Casing	2	400	800
2	3D Printed Impeller	2	420	840
	3D Printed Stator	1	500	500
3	BLDC Motor	2	350	700
4	Butane Fuel Can	4	150	600
5	Fuel Control Valve	1	250	250
6	Fuel Pipe	1(2 foot)	100	100
7	12V LiPo Battery	1	2300	2300
8	LiPo battery Charger	1	450	450
9	Speed Controller	1	150	150
10	30A ESC	1	350	350
11	High Voltage generator	1	300	300
12	Power Switch	1	60	60
13	Stainless Steel Pipes	4	1000	1000
14	Fabrication	5 days	500 (Per Day Cost)	2500
Total Cost				13600/-

Chapter 5: Conclusion and Future Work

This chapter presents the conclusions drawn after working and understanding on the project work.

5.1 Conclusion

- Air Flow Simulation of Jet Engine shows the required swirling motion.
- Area of Stator is **1061.1996** mm^2 & number of slots required is **14**.
- Temperature at Nozzle exit ranges between **270-300 °C.**
- The Jet Model weighs a total of **1,271 grams** or **1.2kg**.
- Temperatures may go up from **500 to 700** ° C during combustion.
- Diameter of holes in the primary, secondary and tertiary hole zone of the combustion chamber are **3.5mm, 5mm & 8mm**
- In summary, the culmination of a jet engine project's success hinges upon delivering a reliable, high-performance engine that not only meets but exceeds specified standards.
- Through meticulous design, exhaustive testing, and innovative engineering, achieving these goals signifies not just a victory for the project team, but also a substantial leap forward for aviation, promising safer, more efficient, and advanced air travel for future generations.

5.2 Future Work

The future work in the field of jet engine development is likely to focus on several key areas to address challenges, improve efficiency, and meet evolving industry demands. Here are some potential avenues for future work in jet engine projects. Develop engines that are more fuel-efficient to reduce environmental impact and operational costs. Explore alternative fuels, such as biofuels or synthetic fuels, to reduce reliance on traditional aviation fuels. Work on technologies to minimize emissions, including the reduction of greenhouse gases and other pollutants produced during combustion.

Investigate advanced combustion processes and emission control systems. Continue efforts to design quieter engines to address concerns related to noise pollution. Research innovative noise reduction technologies and materials. Explore the use of lightweight and high-temperature materials to enhance engine performance. Investigate advanced manufacturing techniques, such as additive manufacturing (3D printing), for more efficient and cost-effective production.

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Additionally, I acknowledge the contributions of my fellow group members, whose collaboration and dedication were crucial to the success of this project.